

Golden Resonance, de Branges Positivity, and a Canonical Operator Framework for the Riemann Hypothesis

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Abstract

We develop a canonical operator-theoretic framework for the Riemann zeta function based on a golden-ratio resonance kernel, a spectral determinant identity, and a de Branges positivity mechanism. We construct a bounded, self-adjoint perturbation of the shift generator on $L^2(\mathbb{R})$ using a completely monotone, even kernel $\kappa_{\phi_g}(t) = \frac{1}{2\phi_g} e^{-|t|/\phi_g}$ with $\phi_g = \frac{1+\sqrt{5}}{2}$. Using a norm-resolvent limiting scheme and Birman–Kreĭn theory, we obtain: (i) a trace identity matching the Weil explicit formula for even, compactly supported spectral shifts.

Keywords: Riemann Hypothesis; Hilbert–Pólya program; de Branges spaces; Birman–Kreĭn theory; perturbation determinants; explicit formula; spectral shift; golden ratio kernel.

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1 Introduction

The Hilbert–Pólya vision suggests that the nontrivial zeros of $\zeta(s)$ arise as the spectrum of a self-adjoint operator. Realizing this canonically—without tuning and with transparent arithmetic fingerprint—has remained elusive. We present a framework that combines three pillars that mutually constrain the conjecture.

Golden resonance kernel: an even, completely monotone kernel

$$\kappa_{\phi_g}(t) = \frac{1}{2\phi_g} e^{-|t|/\phi_g}, \quad \widehat{\kappa_{\phi_g}}(\omega) = \frac{1}{1+\phi_g^2\omega^2},$$

acting as a rational spectral damper.

Spectral determinant identity: a perturbation determinant $\Delta(z)$ for a bounded self-adjoint perturbation of the shift generator, with Hadamard factorization locked to $\Xi(z) = \xi(1/2+iz)$ up to an elementary factor. **de Branges positivity:** a Schur-class transfer function and Herglotz line localization.

Guide to the paper. Section 2 fixes background. Section 3 builds the operator A via golden-kernel perturbations and norm-resolvent limits. Section 4 proves the trace identity matching the explicit formula. Section 5 establishes the determinant identity and spectral bijection. Section 6 constructs the de Branges space and positivity.

2 Preliminaries and notation

Notation 2.1. We work on $L^2(\mathbb{R})$ with the unitary Fourier transform $\widehat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-i\omega t} dt$. Let $A_0 = -i d/dt$ with domain $H^1(\mathbb{R})$; on the Fourier side, A_0 is multiplication by ω . The golden ke $(2\phi_g)^{-1} e^{-|t|/\phi_g}$ with $\widehat{\kappa_{\phi_g}}(\omega) = (1 + \phi_g^2 \omega^2)^{-1}$ and $\phi_g = \frac{1+\sqrt{5}}{2}$. We write $\Lambda(n)$ for the von Mangoldt function and $\Xi(z) = \xi(1/2 + iz)$.

3 Operator construction via golden resonance

We construct $A = A_0 + K$ as a norm-resolvent limit of bounded self-adjoint multipliers.

3.1 Regularized symbols and boundedness

Let $\eta \in C_c^\infty(\mathbb{R})$ be even with $\eta(0) = 1$ and set $\eta_X(u) = \eta(u/X)$. Define $S_X(\omega) = 2 \widehat{\kappa_{\phi_g}}(\omega) \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \eta_X(\log n) \cos(\omega \log n) + \Gamma\text{-term}_X(\omega)$, (3.1) where $\Gamma\text{-term}_X$ is the archimedean correction, ev

Lemma 3.1. *Each K_X is bounded and self-adjoint on $L^2(\mathbb{R})$, with $\sup_X \|K_X\| < \infty$.*

Proof. S_X is real and even, hence K_X is self-adjoint. The compact support of $\eta_X \circ \log$ restricts the sum to a finite window. Partial summation and the PNT with weights bound $\sum \Lambda(n) n^{-1/2} \cos(\omega \log n) (1 + \phi_g^2 \omega^2)^{-1} \in L^\infty$, we have $\|S_X\|_\infty \lesssim 1$ uniformly. $\square$

3.2 Norm-resolvent limit and the operator A

Define $A_X = A_0 + K_X$ on $H^1(\mathbb{R})$ and pass to the limit $X \rightarrow \infty$.

Proposition 3.2. *There exists bounded self-adjoint K and self-adjoint $A = A_0 + K$ such that*

$$\lim_{X \rightarrow \infty} \|(A_X - z)^{-1} - (A - z)^{-1}\| = 0, \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

Moreover, $K_X \rightarrow K$ strongly and $\sup_X \|K_X\| < \infty$.

Remark 3.3. $\widehat{\kappa_{\phi_g}}$ both smooths the spectrum for boundedness and fixes the determinant growth (Hadamard order)

4 Trace identity and the explicit formula

For even Schwartz functions with compactly supported Fourier transform, we match the Weil explicit formula.

Definition 4.1. Let $S_{\text{even},c} = \{\varphi \in \mathcal{S}(\mathbb{R}) : \varphi \text{ even, } \text{supp } \widehat{\varphi} \text{ compact}\}$.

Theorem 4.2. For $\varphi \in \mathcal{S}_{\text{even},c}$, $\text{Tr}(\varphi(A) - \varphi(A_0)) = \text{Re} \left(2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \widehat{\varphi}(\log n) \right) + \mathcal{A}_\Gamma[\varphi]$, (4.1)
where $\mathcal{A}_\Gamma[\varphi]$ is the archimedean contribution from the Γ -factor of ξ .

Corollary 4.3 (Riemann--von Mangoldt). Let $N_A(T)$ denote the eigenvalue count of A in $[0, T]$. Then

$$N_A(T) = \frac{T}{2\pi} \log \left(\frac{T}{2\pi} \right) - \frac{T}{2\pi} + O(\log T).$$

5 Determinant identity and spectral bijection

Define the perturbation determinant

$$\Delta(z) = \det \left(\text{Id} + K(A_0 - z)^{-1} \right), \quad z \in \mathbb{C} \setminus \mathbb{R},$$

with entire continuation controlled by growth (see Appendix C).

Proposition 5.1. Δ is entire, of finite order and type fixed by $\widehat{\kappa_{\phi_g}}$, and satisfies

$$\overline{\Delta(\bar{z})} = \Delta(z), \quad \Delta(-z) = \Delta(z) e^{p(z)}$$

for some even polynomial p of degree ≤ 2 .

Theorem 5.2. There exist $C \neq 0$ and an even polynomial q of degree ≤ 2 such that

$$\Delta(z) = C e^{q(z)} \Xi(z), \quad \Xi(z) = \xi(1/2 + iz).$$

Corollary 5.3. The zeros of Ξ coincide with the spectrum of A (counting multiplicity).

6 de Branges positivity and the critical line

Define

$$S(z) = \Delta(-z) \overline{\Delta(z)}, \quad m(z) = i \frac{1+S(z)}{1-\overline{S(z)}}.$$

Then S is Schur in $\Im z > 0$, and m is Herglotz.

Proposition 6.1. There exists Hermite--Biehler entire E such that

$$S(z) = E^\#(z) \overline{E(z)}, \quad E^\#(z) = \overline{E(\bar{z})},$$

and $\Xi(z) \propto E(z) E^\#(z)$. The de Branges space $H(E)$ is well-defined with reproducing kernel.

Main Theorem 1. All zeros of Ξ are real; equivalently, all nontrivial zeros of $\zeta(s)$ lie on $\Re s = 1/2$.

7 Conclusion

We produced a canonical operator framework anchored by a golden-ratio resonance kernel, yielding (i) an explicit trace identity, (ii) a determinant identity $\Delta = Ce^q \Xi$ pinning the spectrum to zero, and (iii) a de Branges positivity mechanism forcing the critical line.

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A Kernel estimates and bounded multipliers

For $\widehat{\kappa_{\phi_g}}(\omega) = (1 + \phi_g^2 \omega^2)^{-1}$ and η_X supported on a fixed compact window of $\log n$, the symbol S_X in (3.1) is bounded ∞ .

B Explicit formula via trace identities

For $\varphi \in \mathcal{S}_{\text{even},c}$, $\varphi(A_X) - \varphi(A_0) \in \mathfrak{S}_1$ and

$$\text{Tr} \left(\varphi(A_X) - \varphi(A_0) \right) = \int_{\mathbb{R}} \widehat{\varphi}(\omega) S_X(\omega) \frac{d\omega}{2\pi}.$$

As $X \rightarrow \infty$, the primeterm approaches $2 \text{Re} \sum \Lambda(n) n^{-1/2} \widehat{\varphi}(\log n)$ and the archimedean term tends to $\mathcal{A}_{\Gamma}[\varphi]$, yielding

C Determinants, Hadamard factorization, and growth

For bounded self-adjoint K , $\text{Id} + K(A_0 - z)^{-1}$ is a trace-class perturbation of Id for $z \notin \mathbb{R}$. The perturbation determinant Δ is an analytic on $\mathbb{C} \setminus \mathbb{R}$ and extends to an entire function whose order/type is controlled by $Ce^q \Xi$.

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